NYC YELLOW TAXI TRIP DATA ANALYSIS

January 2025 Trip-Level Operational Data

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1. Project Objective

This project performs an industry-style exploratory data analysis of NYC Yellow Taxi trip records for January 2025.

The main objectives are to:

- Understand the structure and characteristics of the taxi-trip dataset.
- Identify and investigate missing, duplicate, invalid, and suspicious records.
- Clean and transform the data appropriately.
- Create meaningful analytical features.
- Perform statistical and exploratory analysis.
- Identify temporal, passenger, location, payment, vendor, and route patterns.
- Create meaningful visualizations.
- Answer important business questions using evidence from the data.
- Communicate operational insights and limitations clearly.

The analysis uses **Python, NumPy, Pandas, Matplotlib, and Seaborn**.

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2. Problem Statement

The NYC Yellow Taxi dataset contains trip-level information such as pickup and drop-off times, passenger count, trip distance, pickup and drop-off location IDs, vendors, payment types, and rate codes.

The objective is to analyze this information to understand:

- When taxi demand is highest and lowest.
- What a typical taxi trip looks like.
- Passenger-count patterns.
- Frequently used pickup and drop-off locations.
- Frequently travelled routes.

- Payment and vendor distributions.
- Trip distance and duration patterns.
- Unusual and potentially problematic records.
- Important data-quality limitations.

The analysis is descriptive and focuses only on information represented in the dataset.

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3. Dataset Description

Dataset: yellow_tripdata_2025-01.parquet

Data type: NYC Yellow Taxi trip records

Period: January 2025

File format: Parquet

Initial records: 3,475,226

Final records after cleaning: 3,475,102

Initial columns: 20

Final columns after feature engineering: 30

The dataset contains trip-level operational information including:

- Vendor ID
- Pickup and drop-off timestamps
- Passenger count
- Trip distance
- Rate code
- Store-and-forward indicator
- Pickup and drop-off location IDs
- Payment type
- Additional trip-related fields

Important Dataset Limitation

The dataset does not provide sufficient information to calculate or claim:

- Actual revenue

- Profit
- Driver earnings
- Customer spending
- Customer-level or driver-level performance

Therefore, the analysis focuses on operational trip characteristics rather than financial performance.

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4. Data Quality Findings

Several data-quality checks were performed before cleaning.

Duplicate Records

No completely duplicated rows were found.

Duplicate records: 0

Missing Values

Missing values were identified in fields including:

- passenger_count
- RatecodeID
- store_and_fwd_flag
- congestion_surcharge
- Airport_fee

The affected records were retained because they still contained useful trip-level information.

Passenger Count

The analysis identified:

24,655 zero-passenger trips

No negative passenger counts were found.

Trip Distance

No negative trip distances were found.

However:

90,893 trips had zero distance.

These records were retained for further investigation rather than automatically removed.

Timestamp Quality

There were:

124 records where drop-off occurred before pickup.

These records were considered invalid because the timestamp sequence was impossible.

Trip Duration

The analysis identified:

- **1,927 zero-duration trips**
- **1,235 trips with positive distance but zero duration**
- **20 trips longer than 24 hours**

The extreme duration records were investigated rather than automatically deleted.

Extreme Distance

The dataset contained extremely large reported trip distances.

The maximum reported distance was:

276,423.57 miles

This was treated as a potential data-quality anomaly.

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5. Data Cleaning Methodology

A separate DataFrame called clean_df was created so that the original dataset was preserved.

The following cleaning decisions were applied:

Duplicate Records

No duplicates were found, so no duplicate records were removed.

Timestamp Cleaning

Pickup and drop-off columns were converted to appropriate datetime types.

The 124 records where drop-off occurred before pickup were removed.

Final dataset:

3,475,102 records

Missing Values

Missing values were investigated and retained where appropriate because removing all affected records would unnecessarily discard useful trip information.

Passenger Count

Zero-passenger records were investigated and retained.

No negative passenger counts were found.

Trip Distance

No negative distances were found.

Zero-distance trips were retained because they may represent unusual but potentially meaningful records.

Extreme Durations

Trips longer than 24 hours were identified and investigated but not automatically removed.

Outliers

IQR-based outlier analysis was performed later.

Statistical outliers were treated as flags rather than automatically classified as invalid records.

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6. Feature Engineering

Several analytical features were created to support the analysis.

Trip Duration

trip_duration_minutes

Calculated from pickup and drop-off timestamps.

Pickup Date

pickup_date

Extracted from the pickup timestamp.

Pickup Hour

pickup_hour

Extracted as the hour of the day.

Day of Week

day_of_week

Created using the pickup timestamp.

Weekend Indicator

is_weekend

Classifies trips as weekday or weekend.

Average Speed

average_speed_mph

Calculated using trip distance and trip duration.

Zero-duration trips were handled to prevent infinite speed values from being treated as valid observations.

Distance Category

Trips were classified into:

- Short
- Medium
- Long

The defined categories resulted in:

- Short: **58.98%**
- Medium: **34.28%**
- Long: **6.74%**

Duration Category

Trips were classified as:

- Very Short
- Short
- Medium
- Long

Route

A combined route field was created using:

PULocationID → DOLocationID

This was used for route-frequency analysis.

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7. Statistical Analysis

NumPy was used for meaningful numerical analysis.

Statistics were calculated for:

- Trip distance
- Trip duration
- Passenger count
- Average speed

The analysis included:

- Mean
- Median
- Minimum
- Maximum
- Variance
- Standard deviation
- 25th percentile
- 50th percentile
- 75th percentile
- 90th percentile
- 95th percentile
- 99th percentile

Trip Distance

- Mean: **5.855 miles**
- Median: **1.67 miles**
- 25th percentile: **0.98 miles**
- 75th percentile: **3.10 miles**
- 99th percentile: **19.50 miles**

The large difference between the mean and median indicates a highly skewed distribution influenced by extreme values.

Trip Duration

- Mean: **15.03 minutes**
- Median: **11.70 minutes**
- 25th percentile: **7.28 minutes**
- 75th percentile: **18.33 minutes**
- 99th percentile: **58.92 minutes**

Passenger Count

- Mean: **1.30 passengers**
- Median: **1 passenger**
- Maximum: **9 passengers**

Average Speed

The calculated speed distribution contained extreme values caused by anomalous distance and duration observations. Therefore, the overall average speed should be interpreted cautiously.

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8. Time-Based Analysis

Taxi activity was analyzed by hour and day of the week.

Busiest Hour

6 PM

Number of trips:

267,949

Least Busy Hour

4 AM

Number of trips:

20,031

Busiest Day

Thursday

Number of trips:

614,208

Least Busy Day

Monday

Number of trips:

363,755

Weekday vs Weekend

- Weekday trips: **2,596,123**
- Weekend trips: **878,979**

Approximately:

- **74.71%** weekday
- **25.29%** weekend

This shows that taxi activity was considerably higher on weekdays in the January 2025 dataset.

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9. Trip Distance & Duration Analysis

The analysis compared trip distance and trip duration to understand typical travel patterns.

Distance

Average:

5.855 miles

Median:

1.67 miles

Duration

Average:

15.03 minutes

Median:

11.70 minutes

The median provides a more representative description of a typical trip because extreme observations significantly affect the mean.

A scatter plot of trip distance versus duration was created to investigate their relationship.

The analysis also identified unusual observations such as:

- Very large distances with short durations.
- Positive distances with zero duration.
- Very short distances with unusually long durations.

These observations were treated as potential data-quality anomalies.

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10. Passenger Analysis

Passenger count was analyzed using frequency distributions and comparisons of distance and duration.

Most Common Passenger Count

1 passenger

Number of one-passenger trips:

2,322,429

Trips with available passenger information:

2,935,071

Percentage of one-passenger trips:

79.13%

One-passenger trips therefore represented the dominant passenger pattern in the dataset.

Passenger distance and duration were also compared across different passenger counts and between weekdays and weekends.

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11. Pickup & Drop-off Location Analysis

The dataset's PULocationID and DOLocationID values represent **TLC Taxi Zone IDs**.

They were therefore analyzed as location IDs rather than neighborhood names.

Top Pickup Location

Location ID 161

Trips:

169,977

Top Drop-off Location

Location ID 236

Trips:

161,376

Most Frequently Travelled Route

237 → 236

Trips:

24,839

Other frequently observed routes included:

- 236 → 237
- 236 → 236
- 237 → 237
- 161 → 237
- 237 → 161
- 161 → 236

Route-level analysis was also performed using a minimum threshold of 100 trips before comparing average duration.

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12. Payment Type Analysis

Payment categories were analyzed using trip counts, percentages, distance, and duration.

Most Common Payment Type

Payment Type 1

Trips:

2,444,387

Percentage:

70.34%

Other payment categories were considerably less frequent.

The analysis treated the missing/zero-coded group cautiously rather than assigning an unsupported payment-method name.

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13. Vendor Analysis

Vendor activity was compared using trip counts, percentages, distance, duration, passenger characteristics, and payment distributions.

Vendor Distribution

Vendor 2

- Trips: 2,719,860
- Percentage: 78.27%

Vendor 1

- Trips: 753,665
- Percentage: 21.69%

Smaller vendor categories were also observed.

The analysis describes differences between vendors but does not classify one vendor as "better" because the dataset does not provide an appropriate overall performance metric.

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14. Rate Code Analysis

Rate codes were analyzed by frequency, percentage, average distance, and average duration.

Most Common Rate Code

Rate Code 1

Trips:

2,756,471

Percentage:

79.32%

Missing rate-code values accounted for approximately **15.54%** of the dataset.

Other rate codes occurred at substantially lower frequencies.

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15. Store & Forward Analysis

The store_and_fwd_flag field was analyzed to compare trip activity.

The majority of observed records were:

N — approximately 84.24%

Trips marked Y represented approximately:

0.22%

The remaining records were missing this value.

Average distance and duration were compared between available categories. These differences are descriptive and should not be interpreted as evidence of causation.

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16. Correlation Analysis

A correlation matrix was created for:

- Passenger count
- Trip distance
- Trip duration
- Average speed

The strongest observed relationship was between:

Trip distance and average speed: approximately 0.785

Other relationships were weak.

For example:

- Passenger count vs duration: approximately **0.023**
- Passenger count vs distance: approximately **0.004**
- Distance vs duration: approximately **0.003**

Correlation measures statistical association and does not establish causation.

The correlation results were visualized using a Seaborn heatmap.

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17. Advanced Analysis

Several advanced analyses were performed.

Demand Heatmap

A day-of-week × pickup-hour heatmap was created to identify periods of high and low demand.

Top 20 Routes

The 20 most frequently observed routes were identified.

The highest-frequency route was:

237 → 236 — 24,839 trips

Route Duration Analysis

Routes with at least 100 trips were compared by average and median duration.

This minimum threshold reduces the risk of drawing conclusions from extremely small groups.

Speed Anomaly Analysis

Extremely high calculated speeds were investigated.

Several observations contained unusually large reported distances combined with very short durations, producing unrealistic speed values.

Distance Categories

- Short: **58.98%**
- Medium: **34.28%**
- Long: **6.74%**

Peak vs Off-Peak

Peak hours were defined as:

- 7 AM–9 AM
- 4 PM–7 PM

Peak-period trips accounted for approximately:

38.74%

Off-peak trips accounted for:

61.26%

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18. Key Findings

The major findings from the analysis are:

1. The final cleaned dataset contains **3,475,102 trips**.
2. Taxi demand was highest at **6 PM**.
3. Taxi demand was lowest at **4 AM**.

4. **Thursday** was the busiest day.
5. Weekdays represented approximately **74.71%** of trips.
6. The median trip distance was **1.67 miles**.
7. The median trip duration was **11.70 minutes**.
8. One-passenger trips represented approximately **79.13%** of trips with available passenger information.
9. Short-distance trips accounted for approximately **58.98%** of all trips.
10. Location ID **161** was the most common pickup location.
11. Location ID **236** was the most common drop-off location.
12. Route **237 → 236** was the most frequently observed route.
13. Payment Type **1** was the most common payment category.
14. Rate Code **1** was the dominant rate code.
15. Several data-quality anomalies were identified, including zero-distance, zero-duration, extreme-distance, and extreme-speed records.

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19. Business / Operational Insights

The analysis provides several operational insights from the January 2025 taxi data.

- Taxi activity was concentrated during afternoon and evening hours, with **6 PM** representing the busiest pickup hour.
- **Thursday** had the highest daily trip volume.
- Weekday demand was substantially higher than weekend demand.
- One-passenger trips represented the dominant passenger pattern.
- Short-distance trips represented the largest distance category.
- Location IDs **161** and **236** were the most frequently observed pickup and drop-off locations respectively.
- Route **237 → 236** had the highest recorded trip frequency.
- The identified demand patterns could help analysts investigate operational resource allocation during high-demand periods.
- Unusual distance, duration, and speed observations should be investigated before being used for operational performance decisions.

- Route-level comparisons should be interpreted carefully when the number of observations is small.
- The analysis describes observed patterns and does not establish causal relationships.

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20. Limitations

The analysis has several important limitations:

- The dataset is a **trip-level operational dataset** and does not provide sufficient information to calculate actual revenue, profit, driver earnings, or customer spending.
- PULocationID and DOLocationID are TLC Taxi Zone IDs rather than neighborhood names.
- A separate lookup table would be required to translate the IDs into neighborhood names.
- Several fields contain missing values.
- Zero-distance and zero-duration trips were identified and retained for investigation rather than automatically removed.
- Extreme distance and duration values can strongly influence averages.
- Calculated average speed is affected by anomalous observations.
- IQR outliers were treated as statistical flags rather than automatically removed.
- Some route and categorical groups contain relatively few observations.
- Correlation does not establish causation.
- The analysis covers **January 2025 only**, so the results may not represent other months or seasons.

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21. Conclusion

The January 2025 NYC Yellow Taxi analysis demonstrates an end-to-end Python exploratory data analysis workflow.

The project covered:

- Data understanding
- Data-quality assessment

- Data cleaning
- Feature engineering
- Statistical analysis
- Exploratory data analysis
- Visualization
- Business questions
- Operational insights
- Limitations
- Final conclusions